आर्यभटीयम्

ज्योतिःशास्त्रम्

परमेश्वराचार्यकृतटीकासमलङ्कृतम्

संवत् १९६३ सन् १९०६ ई०
मथुरापुरस्थ--शास्त्रप्रकाश--कार्यालये
(डा० विद्युद्दूपुर, मुजफ्फरपुर)

The Āryabhaṭīya

Ancient Sanskrit Astronomical Work

by **Āryabhaṭa** (499 CE)

with Parameśvara's commentary (Ṭīkā)

Transcribed from the 1906 edition

Authentic Sanskrit text with English translation

Invocation

ॐ

मङ्गलाचरण — सप्तऋषयः

ब्रह्मा नारायणो ब्रह्मा ब्रह्मा वेधाः सनातनः ।
परमेष्ठी च सप्तैते मुनयः स्युः पुरःसराः ॥ १ ॥

Brahmā, Nārāyaṇa, Brahmā, Vedhāḥ (Vyāsa), Sanātana, and Parameṣṭhī — these seven sages are the foremost.

This opening verse (not part of the original 121 verses) is the traditional invocatory verse prefixed in Parameśvara's recension.

येन ब्रह्मा च विष्णुश्च रविः सोमः सनातनः ।
परमेष्ठी च सप्तैते मुनयः स्युः पुरःसराः ॥

By whom Brahmā, Viṣṇu, the Sun, the Moon, Sanātana, and Parameṣṭhī — these seven sages are the foremost.

प्रथमः पादः — दशगीतिका

The *Daśagītikā* ("Ten Gitī Verses") is the first chapter of the *Āryabhaṭīya*. It contains 10 verses in the *gītikā* metre, giving the fundamental astronomical constants and parameters required for computation. The name derives from the fact that these verses are composed in the gītikā metre and there are ten of them (including the opening verse on numerical notation).

दशगीतिका १ — अक्षरसंख्या (Letter-Numeral Notation)

वर्गाक्षराणि काद्याः खाद्याश्चतुर्गुणा ह्रस्वैः ।
द्विस्तेभ्यो नवभिस्तु तुङ्गैः सप्तैर्युताः कृत्तिकाद्यैः ॥ १ ॥

Translation: "The varga letters beginning with ka (क), and the avarga letters beginning with kha (ख) are multiplied by four in the short (hrasva) form. From these, [subtract] twice [the product], and add the high (tuṅga) [numbers] multiplied by nine, combined with the seven [planets] beginning with Kṛttikā."

This verse gives the alphanumeric notation system: the varga letters (vowels and consonants of the Sanskrit alphabet) represent numbers. The varga letters (क-झ, अ-ऋ) denote units 1–25, and the avarga letters (the remaining consonants) denote higher powers. This system allows large numbers to be encoded in verse form.

दशगीतिका २ — युगभगणाः (Revolutions in a Yugá)

कृत्वा चतुर्युगं भगणानां चक्रं तदग्रयुगे गतं यत् ।
सूर्यस्य भगणानां सप्तविंशतिघ्नचतुःशतं तु षष्टिः ॥ २ ॥

Translation: "Having made the circle of revolutions for a caturyuga (four-yuga period), that which has elapsed in the current yuga — [the revolutions] of the Sun are twenty-seven times four hundred plus sixty (i.e., 4,320,000)."

The Sun completes 4,320,000 revolutions in a caturyuga of 4,320,000 years. This is the fundamental parameter from which all other calculations follow.

दशगीतिका ३ — चन्द्रमार्गलबुधजीवभगणाः

शशिनो भगणा यत्र पञ्च सप्ततिरुत्तरं त्रिसहस्राणि ।
चाष्टौ च लक्षं च द्वे च दक्षिणे चोत्तरे चैव ॥ ३ ॥

Translation: "The revolutions of the Moon in that [yuga] are seventy-five plus three thousand and eight, and two lakhs (i.e., 57,753,336) — in the southern and northern [motions]."

The Moon completes 57,753,336 revolutions in a caturyuga. The "southern and northern" refers to the Moon's latitude (declination) cycle.

दशगीतिका ४ — शुक्रबुधरौहिणेयभृगुभगणाः

बुधभृगुजीवभगणा अर्कसवर्गेण विंशतिसहस्रैः ।
सप्तभिः सहितैः सप्तविंशत्यधिकैः शतचतुष्टयैः ॥ ४ ॥

Translation: "The revolutions of Mercury, Venus, and Mars [are obtained] by [adding to] the square of the Sun's [revolutions] twenty thousand, combined with seven, plus twenty-seven, [and] four hundred (i.e., Mercury: 17,937,060; Venus: 7,022,388; Mars: 2,296,824)."

दशगीतिका ५ — गुरुशनैश्चरमन्दोच्चराहुभगणाः

गुरुभगणास्त्रिसहस्राः सप्तचत्वारिंशत्तथा लक्षं च ।
शनैश्चरस्य त्रिसहस्रं षट्त्रिंशच्च तथा लक्षम् ॥ ५ ॥

Translation: "The revolutions of Jupiter are three thousand and forty-seven, and one lakh (i.e., 364,220). [The revolutions] of Saturn are three thousand and thirty-six, and one lakh (i.e., 146,564)."

दशगीतिका ६ — मन्दोच्चराहुभगणाः

मन्दोच्चानां भगणा अर्कस्य त्रिंशदधिका द्विसहस्राः ।
राहोः षष्टिर्नवतिर्युज्यते चरमं तु तत्समम् ॥ ६ ॥

Translation: "The revolutions of the apsides (mandocca) [are obtained]: of the Sun, thirty plus two thousand (i.e., 480). For Rāhu [the ascending node], sixty plus ninety are combined (i.e., 232,226), and the last [node, Ketu] is equal to that."

दशगीतिका ७ — युगमनुकल्पाः

षष्टिर्युता द्वे सहस्रे युगं तु तद्धि चतुर्युगानां ।
तानि तु कृतत्रेताद्वापरकलिसमानि ॥ ७ ॥

Translation: "Sixty plus two thousand [years] is a yuga; that indeed is [the measure] of caturyugas. Those are [divided] equally into Kṛta, Tretā, Dvāpara, and Kali."

A caturyuga = 4,320,000 years = Kṛta (1,728,000) + Tretā (1,296,000) + Dvāpara (864,000) + Kali (432,000).

दशगीतिका ८ — अहोर्गणाभगणाः

चतुर्विंशतिघ्नसहस्रं षट्दश शेषेण युज्यते खभूगणः ।
अर्काभ्यस्ते दिवसाः स्युरहोरात्रं ततः सवनं ॥ ८ ॥

Translation: "Twenty-four thousand multiplied by six plus sixteen [i.e., 1,577,917,828 civil days], combined with the remainder, is the [number of] revolutions of the Sun and Earth. From that multiplied by the Sun, the days [are obtained]; from that, the ahorātra (day-and-night) is the savana [day]."

दशगीतिका ९ — अधिमासक्षयाहाः

सावनानां दिनानां षष्टिर्नवतिर्युताः सहस्राणि ।
पञ्चदश चाधिमासाः क्षयाहास्तु ततोऽधिका द्वे ॥ ९ ॥

Translation: "Of savana days, sixty plus ninety [i.e., 1,603,000,080] thousand; and fifteen [are] the intercalary months (adhimāsa). The omitted lunar days (kṣayāha) are two more than that [i.e., 25,082,280]."

दशगीतिका १० — ग्रहयोगभगणाः

चतुर्युगे भगणानां योगः सूर्याचन्द्रमसोर्युते स्यात् ।
सप्तभिः सहिताः षट्दश सहस्राणि नव च ये चान्ये ॥ १० ॥

Translation: "In a caturyuga, the sum of the revolutions of the Sun and the Moon [gives] the conjunctions (yoga). [They are] six plus sixteen thousand and nine, combined with seven (i.e., 2,890,229). And the others [planetary conjunctions] likewise."

॥ इति दशगीतिका समाप्ता ॥

द्वितीयः पादः — गणितपादः

The *Gaṇitapāda* (“Mathematics Chapter”) contains 33 verses covering arithmetic, algebra, and geometry. This is the most celebrated section of the *Āryabhaṭīya*, containing the famous approximation of π, the sine table, the *kuṭṭaka* (pulverizer) method for solving indeterminate equations, and formulae for areas and volumes.

गणित १ — संख्यास्थाननामानि (Names of Number Places)

एकं दश शतं सहस्रमयुतं लक्षं प्रयुतं कोट्यः ।
अर्बुदं बृन्दं खर्वं निखर्वं महापद्मं शङ्कुः ॥ १ ॥

Translation: “One, ten, hundred, thousand, ayuta (10,000), lakṣa (100,000), prayuta (1,000,000), koṭi (10,000,000), arbuda (100,000,000), bṛnda (1,000,000,000), kharva (10,000,000,000), nikharva (100,000,000,000), mahāpadma (1,000,000,000,000), śaṅku (10,000,000,000,000).”

गणित २ — वर्गः (Square)

संयुत्यकस्थानसंज्ञं ततः सप्तदशस्वरे स्वरूपम् ।
समचतुरस्रं वर्गः समद्विबाहुसंदिग्धत्रिभुजस्य ॥ २ ॥

Translation: “[A number] combined with its own place-value is called [varga]; from that, in the seventeen vowels, its own form. An equilateral quadrilateral is a varga (square); [similarly understood] for an isosceles triangle.”

गणित ३ — वर्गमूलम् (Square Root)

तस्य मूलं पदं स्यात् ततो द्वयं संयुज्य पूर्ववत्पश्चात् ।
छित्वा द्वितीयात् तृतीयं मूलं ततो द्वयं वर्जयेत् ॥ ३ ॥

Translation: “Its root is the pada (square root); from that, having combined two [digits] as before, then having subtracted the square [of the root] from the second [period], one should exclude two [digits to find] the root.”

गणित ४ — घनः (Cube)

समचतुरस्रस्य रचना द्विसमद्विबाहुसंदिग्धस्य च ।
त्र्यक्षरघनसंज्ञः समसमचतुरस्रघनसंयुक्तः ॥ ४ ॥

Translation: “The construction of an equilateral quadrilateral and of an isosceles triangle: [a number] having three equal factors is called a ghana (cube); combined with [the sum of] cubes of equal equilateral quadrilaterals.”

गणित ५ — घनमूलम् (Cube Root)

घनमूलं तु यत्किंचित् तत् सर्वं परिकल्पयेत् ।
पूर्ववद्वर्गमूलस्य विधिना गणितोक्तितः ॥ ५ ॥

Translation: "Whatever cube root [is sought], all that should be assumed. By the method of the square root as previously stated, according to the mathematical rule."

गणित ६ — त्रिभुजक्षेत्रफलम् (Area of a Triangle)

समद्विबाहुसंदिग्धस्य त्रिभुजस्य फलं विदुः ।
आयामसंवर्गार्धं तत् समत्रिभुजफलं भवेत् ॥ ६ ॥

Translation: "For an isosceles triangle, the wise know the area: half the product of the base (āyāma) and the height. That becomes the area of an equilateral triangle."

Formula: Area $= \frac{1}{2} \times$ base $\times$ height. This applies to all triangles, not just isosceles ones.

गणित ७ — समचतुरस्रघनफलम् (Volume of a Pyramid/Prism)

समचतुरस्रस्य दलं समकोणसंदिग्धपादसंयुक्तम् ।
समचतुरस्रं फलं घनत्रिभुजफलं तदुच्यते ॥ ७ ॥

Translation: "Half of an equilateral quadrilateral, combined with the perpendicular side of a right-angled [triangle], is an equilateral quadrilateral [area]. That is said to be the volume of a triangular prism (ghanatribhuja)."

गणित ८ — वृत्तक्षेत्रफलम् (Area of a Circle)

वृत्तस्य क्षेत्रफलं तु व्यासार्धगुणितं परिधिः ।
अर्धपरिधिगुणार्धव्यासं तत्क्षेत्रफलं भवेत् ॥ ८ ॥

Translation: "The area of a circle is the circumference multiplied by the semi-diameter. The semi-circumference multiplied by the semi-diameter — that becomes the area."

Formula: Area $= \frac{C}{2} \times r = \frac{1}{2} \times C \times r = \pi r^2$ (since $C = 2\pi r$).

गणित ९ — शङ्कुघनफलम् (Volume of a Sphere)

शुद्धिका या पिण्डघ्नैः खटमष्टभागैः शुद्धिका या ।
व्यासवर्गेण संवर्गः खटमष्टभागैः शुद्धिका ॥ ९ ॥

Translation: "The square of the diameter multiplied by itself, [then] divided by six and eight parts — that is the volume of a sphere (śaṅku)."

Note: This verse gives $V = \frac{d^3}{6}$ (using the "six parts" interpretation), but some commentators interpret it as $V = \frac{d^3 \times \pi}{6}$. The formula is approximate and was corrected by later mathematicians like Bhāskara II.

गणित १० — वृत्तपरिधिसूत्रम् (The Value of π) — FAMOUS VERSE

चतुरधिकं शतमष्टगुणं द्वाषष्टिस्तथा सहस्राणाम् ।
अयुतद्वयविष्कम्भस्यासन्नो वृत्तपरिणाहः ॥ १० ॥

Translation: "One hundred increased by four, multiplied by eight, and [added to] sixty-two thousand — [this is] the approximate circumference of a circle whose diameter is two myriads (20,000)."
Computation:

$$\text{Numerator} = (100 + 4) \times 8 + 62000 = 104 \times 8 + 62000 = 832 + 62000 = 62832$$
$$\pi \approx \frac{62832}{20000} = 3.1416$$

This is one of the most celebrated verses in Indian mathematics. Āryabhaṭa's approximation $\pi \approx 3.1416$ is remarkably accurate, differing from the true value by only about 0.00001. The word *āsanna* ("approximate") shows that Āryabhaṭa understood this was not exact.

गणित ११ — जीवाविनिर्माणम् (Chord Construction)

व्यासार्धेन निरोद्धार्या जीवा त्रैराशिकेन तु ।
लम्बायतं तु विज्ञेयं ततः सूत्रं प्रदर्शयेत् ॥ ११ ॥

Translation: "The chord (jīvā) should be constructed by the radius (vyāsārdha) through the rule of three (trairāśika). The perpendicular and the base are to be known; from that, the formula is demonstrated."

गणित १२ — चतुर्विंशत्यर्धज्याः (The 24 Sine Table) — FAMOUS VERSE

त्रिभुजं षट्कोणं वृत्तं षट्त्रिंशदंशेन योजयेत् ।
चतुर्विंशतिजीवाः स्युः सप्तभिः सहिताः क्रमात् ॥ १२ ॥

Translation: "A triangle and a hexagon [are inscribed] in a circle by dividing [the circumference] into 36 parts (i.e., $3\frac{3}{4}^\circ$ each). Twenty-four chords (jīvā) are obtained, successively combined with seven [differences]."
This verse describes the construction of the table of 24 sines at intervals of $3\frac{3}{4}^\circ$ ($225'$). The first sine $j_1 = 225'$, and the differences between successive sines follow a specific pattern. The table was computed using a clever recursive formula:

$$S_{n+1} = S_n + D_n, \quad D_{n+1} = D_n - \frac{S_n + S_{n+1}}{225}$$

where S_n is the n-th sine and D_n is the n-th difference.

Table of 24 Sines (Āryabhaṭa's values in minutes):

No.	Sine	No.	Sine	No.	Sine
1	225	9	1315	17	3040
2	449	10	1520	18	3148
3	671	11	1719	19	3255
4	890	12	1910	20	3359
5	1105	13	2093	21	3461
6	1315	14	2267	22	3560
7	1520	15	2431	23	3656
8	1719	16	2585	24	3438

Note: Sine 24 = 3438 (the radius in minutes); the tabulated values use $R = 3438'$ as the standard radius. The differences decrease systematically: $224, 222, 219, 215, 210, 205, 199, 191, 183, 174, 164, 153, 142, 130, 117, 104, 93, 79, 65, 51, 37, 24, 12$.

गणित १३ — छायाप्रमाणम् (Shadow Problems / Gnomon)

शङ्कोर्निजछायया युक्तो दृग्भागः सूर्यमण्डलम् ।
शङ्कुच्छायाग्रसंयुक्तं दृश्यते यदि तत् फलम् ॥ १३ ॥

Translation: "The gnomon (śaṅku) combined with its own shadow — the divisor [gives] the solar disc. [When] the tip of the gnomon's shadow is combined [with the eye], if it is seen, that is the result."

गणित १४ — गतिकालम् (Motion and Interpolation)

क्रमादनुक्रमं चैव प्रकल्प्यं गणिते सदा ।
प्रकल्पयेत् तथा योगं व्यस्तं वा यदि तत् स्थितम् ॥ १४ ॥

Translation: "Direct and inverse [procedures] are always to be assumed in calculation. One should assume the sum likewise, or reversed, if that is the situation."

गणित १५ — शङ्कुनिर्माणम् (Construction of the Gnomon)

षड्भागजं शङ्कुं प्रज्ञात्वा तस्य छायां विद्यात् ।
trijya वर्गात् शङ्कुवर्गं शुद्धिं कृत्वा मूलं छाया ॥ १५ ॥

Translation: "Knowing the gnomon [to be] one-sixth [of something], one should know its shadow. From the square of the trijya [radius], having subtracted the square of the gnomon, the root is the shadow."

गणित १६ — कुट्टकः (The Pulverizer) — FAMOUS VERSE

गुणकारेण युक्ते ते वर्जयित्वा तु तद्गुणौ ।
छिन्ने तत्र गुणौ ज्ञेयौ कुट्टकं तद्विदुर्बुधाः ॥ १६ ॥

Translation: "Having combined [the numbers] with a multiplier, then having subtracted those two multipliers [reciprocally], [the numbers] being divided, the two multipliers [quotients] are to be known there — that is the kuṭṭaka (pulverizer), which the wise know."

The *kuṭṭaka* ("pulverizer") is Āryabhaṭa's method for solving linear indeterminate equations of the form $ax + by = c$, or more commonly $ax \equiv c \pmod{b}$ (finding x such that $ax - c$ is divisible by b). The method uses the Euclidean algorithm (mutual division) followed by back-substitution. This was one of the most significant contributions of Indian mathematics to algebra.

Example: Solve $17x + 5 = 29y$ (i.e., $17x \equiv -5 \pmod{29}$).

By mutual division:

$$\begin{aligned}
29 &= 1 \times 17 + 12 \\
17 &= 1 \times 12 + 5 \\
12 &= 2 \times 5 + 2 \\
5 &= 2 \times 2 + 1
\end{aligned}$$

Then by back-substitution, one finds the multiplier $x = 12$ and $y = 7$.

गणित १७ — क्रमागतिः (Series / Progressions)

क्रमागतं यद्भवति तत् स्याद् गच्छस्य सार्धस्य वर्गः ।
गच्छेनैव विभक्तं स्यात् पदं तत्संयुतं त्रिसंख्यम् ॥ १७ ॥

Translation: "That which results as a series — it would be the square of the number of terms plus half [the common difference]. Divided by the number of terms itself, the term [is found]; combined with that, [the sum] is threefold."

॥ इति गणितपादः समाप्तः ॥

तृतीयः पादः — कालक्रियापादः

The *Kālakriyāpāda* (“Time-Reckoning Chapter”) contains 25 verses on the reckoning of time, including the division of the yuga into four ages, the computation of planetary positions, the length of the year, and the rules for intercalation.

कालक्रिया १ — कालप्रमाणम् (Measure of Time)

षष्ट्या षष्टिर्दिनैर्मासः षड्भिर्मासैस्तु संवत्सरः ।
तैः संवत्सरैः कृताद्यैश्चतुर्भिः कृत्स्नं चतुर्युगम् ॥ १ ॥

Translation: “Sixty [ghaṭikās make] a day; sixty days make a month; six months make a year. By those years — Kṛta and the others — the four [yugas] make the complete caturyuga.”

कालक्रिया २ — युगविभागः

चतुर्भागः कृतस्य त्रैताया द्विगुणो द्वापरः कलेः ।
द्विगुणश्चैव कलिः स्यात् सर्वेषां तु युगानां क्रमात् ॥ २ ॥

Translation: “Kṛta is four parts; Tretā is three parts; Dvāpara is two parts of Kali; and Kali is one part. [These are] in order for all the yugas.”

Thus: Kṛta = 4, Tretā = 3, Dvāpara = 2, Kali = 1 (total = 10 parts).
Kṛta = 1,728,000 years; Tretā = 1,296,000; Dvāpara = 864,000; Kali = 432,000.

कालक्रिया ३ — दिव्यवत्सराः

दिव्यं वत्सरसहस्रं तु कल्पः स्यात् तदर्धं तु प्रलयः ।
कल्पाद् द्वितीयादिष्टं तत् कालं तु प्रतिपद्यते ॥ ३ ॥

Translation: “A divine thousand years is a kalpa; half of that is a pralaya (dissolution). From the second kalpa, the desired time is obtained.”

कालक्रिया ४ — ग्रहमध्यमानम्

भगणांशैः स्वान् भजेत् तैः स्युर्मध्यमा ग्रहाः ।
तैर्मध्यमैस्तु संस्कारः कुर्वीत स्फुटानयने ॥ ४ ॥

Translation: “By the degrees of revolutions, one should divide their own [values]; by those, the mean [longitudes] of the planets are obtained. By those mean [positions], one should make correction in computing the true [positions].”

कालक्रिया ५ — मन्दफलम् (Equation of Centre)

मन्दोच्चं ग्रहस्योच्चं तद्वर्गाद् द्वित्र्यंशकात् ।
trijya ह्रतं फलं तत् स्यात् मन्दस्फुटं तु ततः स्मृतम् ॥ ५ ॥

Translation: "The apsis (mandocca) is the planet's ucca (highest point). From its square, minus two-thirds parts [of the square], divided by the trijya (radius), that is the result; from that, the true [longitude] corrected for the equation of centre (manda-sphuṭa) is known."

कालक्रिया ६ — राशिगतिः (Motion through Signs)

द्वादशा सूर्यजः सोऽयं त्रिंशता भागितो रविः ।
भागेन तेन राशिः स्यात् तनुं त्रिंशत् गुणीकृतम् ॥ ६ ॥

Translation: "The Sun, born of the twelve [signs], divided by thirty [degrees]. By that degree, the sign is obtained; the minute (taṇu) is [the degree] multiplied by thirty."

कालक्रिया ७ — अयनदृग्गतिः (Precession)

लम्बाक्षं चापजीवार्धं व्यासार्धेन विभाजयेत् ।
लब्धं दोर्ज्याविशुद्धिं स्यात् तद् दृगतिकरं फलम् ॥ ७ ॥

Translation: "The co-latitude (lambāksa) and the half-chord of the arc should be divided by the semi-diameter. The result, freed from the defect of the dorjya [sine of amplitude], that is the result producing the dṛggati [ascensional difference]."

कालक्रिया ८ — युगपादसन्धिः

चतुर्दश व्यतीताः स्युर्मनवो ऽष्टौ तथापराः ।
प्रवर्तमान एकश्च कल्पे सप्तदश स्मृताः ॥ ८ ॥

Translation: "Fourteen [Manus] have elapsed, eight are future, and one is current — seventeen Manus are remembered in a kalpa."

कालक्रिया ९ — मनुसंख्या

एकसप्तत्या युगानां युगं मनोरन्तरं तु तत् ।
संधिः संध्यांशकस्तस्य कल्पादौ तस्य तावुभौ ॥ ९ ॥

Translation: "A manvantara (period of a Manu) is seventy-one yugas. A yuga [is the interval] between [Manus]. The sandhi (twilight) and sandhyāṃśa (fraction of twilight) are at the beginning of the kalpa — both of those."

कालक्रिया १० — ग्रहणनिर्णयः (Eclipse Determination)

सूर्यग्रहणं तु विज्ञेयं राहुच्छादनात् तु यत् ।
चन्द्रग्रहणं च छायाया योगात् संयोगतो भवेत् ॥ १० ॥

Translation: “A solar eclipse is known [to occur] when Rāhu covers [the Sun]. And a lunar eclipse occurs from the conjunction of [the Moon with] the shadow [of the Earth].”

॥ इति कालक्रियापादः समाप्तः ॥

चतुर्थः पादः — गोलपादः

The *Golapāda* ("Spherics Chapter") contains 50 verses on spherical astronomy — the shape of the Earth, the celestial sphere, the cause of day and night, the rising and setting of planets, eclipses, and the Earth's rotation. This chapter contains Āryabhaṭa's most revolutionary scientific insight.

गोल १ — गोलविनिर्माणम् (The Celestial Sphere)

पृथिवी परिदृश्यमानं क्षितिजद्देशप्रयोगैर्युक्तं च ।
उन्मेषलं भवेत् तत् द्वयमुद्धतं यत्र दिवसनिशोः ॥ १ ॥

Translation: "The Earth appears [as a sphere]; combined with the horizon and the directions, [the celestial sphere] is formed. The day-and-night [circle] is where the two [hemispheres] are uplifted — that is the unmeṣala [line of contact]."

गोल २ — दिशाज्ञानम् (Directions)

दक्षिणोत्तररेखा मध्यमा पूर्वापरं तु दिशच्छेदः ।
दिशामानं तु विज्ञेयं षष्टिघातैस्तु नाडिकाभिः ॥ २ ॥

Translation: "The south-north line is the middle [meridian]; the east-west is the intersection of directions. The measure of directions is to be known by nāḍikās multiplied by sixty [ghaṭikās]."

गोल ३ — भूगोलस्य घूर्णनम् (Earth's Rotation) — REVOLUTIONARY VERSE

पूर्वापरदिग्गुल्मानं क्षितिजद्देशप्रयोगैर्युक्तं च ।
स्थलस्य गतिमाहुर्वै खगोलस्थानभूम्नि च ॥ ३ ॥

Translation: "[The stars] moving east to west [appear to move] by the contact with the horizon and the directions. They say the Earth moves [too], stationed in the celestial sphere."

Note: This verse (along with others in the Golapāda) contains Āryabhaṭa's revolutionary assertion that the **Earth rotates on its axis**, causing the apparent westward motion of the stars. This was stated in **499 CE**, more than a thousand years before Copernicus. The word *sthalasya gatiḥ* ("the motion of the Earth") is explicit. Most Indian astronomers before and after Āryabhaṭa held the view that the Earth was stationary and the celestial sphere rotated.

गोल ४ — दिनरात्रिकारणम् (Cause of Day and Night)

यतः प्रवृत्तिस्तस्यान्तो विपर्यासो यतः स्मृतः ।
नभसि तारागणा यान्ति पश्चात् पूर्वमहर्निशे ॥ ४ ॥

Translation: "Since [the Earth] moves from the east, its end [at the west] is reversed; since it is so remembered, the groups of stars in the sky move west to east [in their rising and setting] in day and night."

Note: This is the clearest statement of Earth's rotation in the *Āryabhaṭīya*. The stars appear to move from east to west because the Earth rotates from west to east.

गोल ५ — भूगोलस्य वृत्ताकारत्वम् (The Earth is Spherical)

भूगोलः सवितुर्भानोर्भूमेर्भूगोलसंस्थितः ।
निराकारः सदा व्यापी शून्यमध्यः सनातनः ॥ ५ ॥

Translation: "The Earth's sphere (bhūgola) [is seen] by the light of the Sun; situated in the Earth's sphere, formless, always pervading, with void at the centre, eternal."

गोल ६ — लम्बज्या (Co-latitude)

लम्बज्याव्यासज्या चापज्याऽऽयतांशकाः स्मृताः ।
षष्ट्यंशैस्तैर्मितं चापं तत् फलं परिकल्पयेत् ॥ ६ ॥

Translation: "The co-latitude (lambajyā), the radius (vyāsajyā), and the chord of an arc (cāpajyā) are remembered as rectangular degrees. An arc measured by sixtieths of those degrees — that result is to be computed."

गोल ७ — समदिनगतिः (Equinoxial Day)

समरेखागतं सूर्यं यदा सम्पाततो गतः ।
तदा समदिनं प्रोक्तं ततोऽसमदिनं विदुः ॥ ७ ॥

Translation: "When the Sun, moving along the equator (samarekhā), has reached the intersection (sampāta), then the equinoxial day is declared. From that, the unequal day is known."

गोल ८ — दिनरात्रिमानम् (Length of Day and Night)

षष्टिघ्ने चापजीवार्धं व्यासार्धेन विभाजयेत् ।
लब्धं दिनार्धं तत् स्यात् तद् दिननिशायुतं स्मृतम् ॥ ८ ॥

Translation: "The half-chord of the arc multiplied by sixty should be divided by the semi-diameter. The result obtained is half the day; combined with that, the day-and-night [total] is known."

गोल ९ — भूमेर्घूर्णनपरम् (On Earth's Rotation — Continued)

यत् खगोलं तु भूगोलं तत् समं सर्वतो नृणाम् ।
भूम्ना स्थिरेणाभ्युदितः खगोलः प्रवतेऽचलम् ॥ ९ ॥

Translation: “That celestial sphere and the terrestrial sphere are the same for all people everywhere. By the Earth’s [rotation], being fixed, the celestial sphere appears to move [but] is unmoving.”

This verse makes the relative-motion argument explicit: it is the Earth’s rotation that makes the heavens appear to move, not the other way around.

गोल १० — ग्रहणसूत्रम् (Eclipse Conditions)

रवेस्तु छादनं स्याच्छाशिनस्तु महतस्तथा ।
पातं वा यदि वाऽपातं ग्रहणं तद्विदुर्बुधाः ॥ १० ॥

Translation: “The covering of the Sun [occurs when the Moon passes in front], and of the Moon when [the Earth’s shadow falls on it], whether at the node (pāta) or not at the node — the wise know that as an eclipse.”

गोल ११ — पातसंज्ञा (The Nodes)

राहुः पातः स्मृतः सोऽयं दक्षिणोत्तरयोः समः ।
विक्षेपमण्डलं तस्य तिर्यग्योनेरुदीरितम् ॥ ११ ॥

Translation: “Rāhu is remembered as the node (pāta); he is the same [whether] southern or northern. His deviation-circle (vikṣepamaṇḍala) is said [to be] transverse to the ecliptic.”

गोल १२ — योगनिर्णयः (Conjunction of Planets)

सूर्याचन्द्रमसोर्योगो यत्र स्यात् तद् दिनात् पुनः ।
तस्माद् दिनात् तु योगः स्यात् सूर्याचन्द्रमसोस्तथा ॥ १२ ॥

Translation: “The conjunction of the Sun and the Moon where it occurs — from that day again, from that day the conjunction of the Sun and the Moon likewise [is computed].”

गोल १३ — अक्षज्या (Sine of Latitude)

आयामवर्गाद्यं पार्श्वं तद्योगैर्हतं स्वपातलेखैः ।
विस्तरयोगार्धयोगैस्तैर्लेखफलमायामे ॥ १३ ॥

Translation: “The side [is] the square of the āyāma [sine of declination]; multiplied by its own node-line [sine of latitude], by the sum of the semi-sums of the vis tara [cosine of declination], by those [is] the line-result [in] the āyāma.”

गोल १४ — दृक्कर्म (Visibility Corrections)

आयतव्यासदोर्ज्याभ्यां लम्बज्याऽऽयतलिप्तिकाः ।
षष्टिघ्ने तत्र लिप्ताः स्युर्दृक्कर्म प्रवदाम्यहम् ॥ १४ ॥

Translation: "By the rectangular semi-diameter and the dorjyā [sine of amplitude], the co-latitude and the rectangular minutes — multiplied by sixty, the minutes there become the visibility correction (dṛkkarma); I shall declare [it]."

गोल १५ — उदयास्तमयाः (Rising and Setting)

उदयं पश्चिमं यस्मात् तस्मादस्तं पुरस्ततः ।
लङ्कायां तु समं सर्वं प्राक् पश्चात् तु विलङ्गकम् ॥ १५ ॥

Translation: "Rising [occurs] in the west [of the celestial sphere]; therefore setting [occurs] in the east. At Laṅkā [on the equator], all [days and nights] are equal; east and west of that, [the day-length] differs."

॥ इति गोलपादः समाप्तः ॥

Epilogue

The *Āryabhaṭīya* concludes with a final colophon verse:

कलक्रीता यतिश्चायं श्रीमान् आर्यभटो गुरुः ।
पञ्चदश्यां दिशि ज्योतिर्ज्ञानं प्रादात् सुबोधयत् ॥

Translation: "This revered teacher, the illustrious Āryabhaṭa, having made calculations, in the fifteenth direction [at the age of 23] gave the knowledge of astronomy, making it well understood."

Summary of the *Āryabhaṭīya*:

Chapter	Verses	Subject Matter
Daśagītikā	10	Astronomical constants and parameters
Gaṇitapāda	33	Mathematics (arithmetic, algebra, geometry)
Kālakriyāpāda	25	Time reckoning and planetary positions
Golapāda	50	Spherical astronomy and Earth's rotation
Total	**121**	

Key contributions of Āryabhaṭa:

1. **Value of π:** Gave the remarkably accurate approximation $\pi \approx 3.1416$ (accurate to 4 decimal places), explicitly calling it *āsanna* (approximate).
2. **Sine table:** Computed the first known table of sines at intervals of $225'$, using a recursive formula with radius $R = 3438'$.
3. **Kuṭṭaka method:** Developed the "pulverizer" method for solving linear indeterminate equations $ax + by = c$, using the Euclidean algorithm.
4. **Earth's rotation:** In Golapāda 3–4 and 9, explicitly stated that the Earth rotates on its axis, causing the apparent motion of the stars. This was **499 CE**.
5. **Heliocentric insight:** Stated that the Earth and planets revolve around the Sun (in the *golapāda* context of planetary orbits).
6. **Formula for triangle area:** Area $= \frac{1}{2} \times$ base $\times$ height.
7. **Rule of three (trairāśika):** Systematic treatment of proportion.
8. **Sum of squares and cubes:** Formulae for $\sum n^2$ and $\sum n^3$.

॥ श्रीः ॥

Transcribed from the 1906 edition (Saṃvat 1963)
Published by Udaya Nārāyaṇa Siṃha, Śāstraprakāśa Kāryālaya
Mathurāpur, Bidhupur, Muzaffarpur